ME623

Dynamics of Machining Processes

Chatter Detection in Turning using Supervised Machine Learning

A PHYSICS-INFORMED LABELING APPROACH

Group 7

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Problem & Methodology Overview

Detecting chatter -- a self-excited tool vibration -- from turning-process cutting-force signals

The problem

Chatter causes poor surface finish, tool wear, and scrap. No expert labels exist for this dataset -- chatter must be inferred from the 3-axis cutting-force signal (10 kHz, 18 turning cuts, 3 spindle speeds, DOC 100-1400 um) itself.

Two contributions in this project

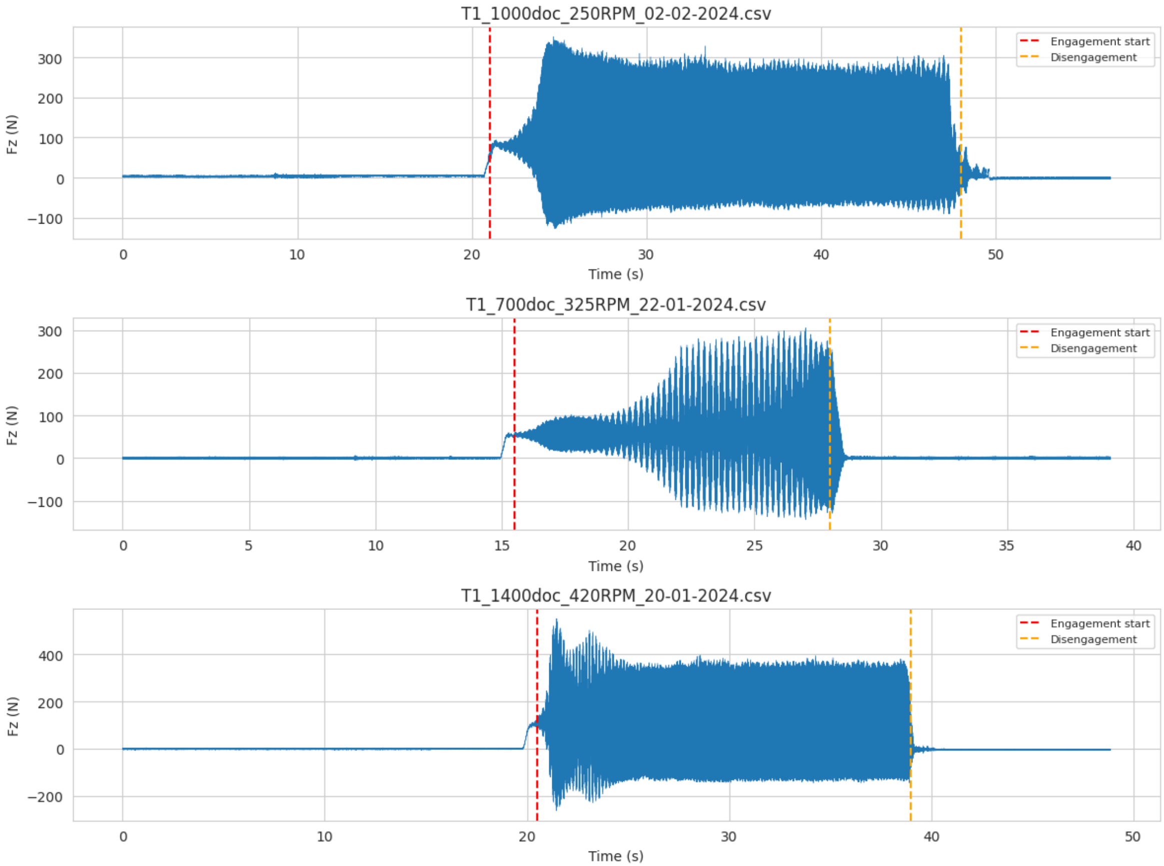
- Automatic engagement detection -- isolating genuine cutting from no-load idle time in every recording
- A physics-informed chatter label, replacing a naive amplitude-percentile cutoff with literature-grounded criteria

Pipeline

1. Engagement detection (+ charge-drift guard)
2. Feature extraction (1s windows, time + freq domain)
3. Ablation study validating engagement detection
4. Literature review -> physics-informed label
5. Feature selection with a leakage check
6. Modeling (RF/XGBoost/SVM/LogReg), F2-based selection
7. Stability diagram for process planning

Automatic Engagement Detection

Every recording includes no-load idle time before/after cutting; idle duration varies 9-25s per file



Method

1. Per-file idle baseline from resultant-force RMS.
2. Flag "engaged" once RMS sustainably exceeds baseline x5+5N.
3. Guard against piezoelectric charge-drift: track the AC/fluctuation component separately from the mean level.
4. 1 of 18 files never engages (100um DOC, below min. chip thickness) -- excluded, not mislabeled.

Ablation study confirms this matters: F1 drops from ~0.87 (idle included) to ~0.44 (engaged only) under an identical pipeline

From the Literature: Forced Vibration vs. Self-Excited Chatter

A turning-specific review motivating a better chatter label than a percentile cutoff

1 Forced vs. self-excited vibration

Chatter is not driven by an external periodic force -- it grows on its own, typically away from spindle-rotation harmonics. Delio, Tlusty & Smith (1992); Kondo, Ota & Kawai (1997).

2 Amplitude ratio as a severity metric

Compare peak amplitude against the expected forced-vibration content in the same spectrum. "Experimental validation of the amplitude ratio as a metric for milling stability identification."

3 Harmonic exclusion criterion

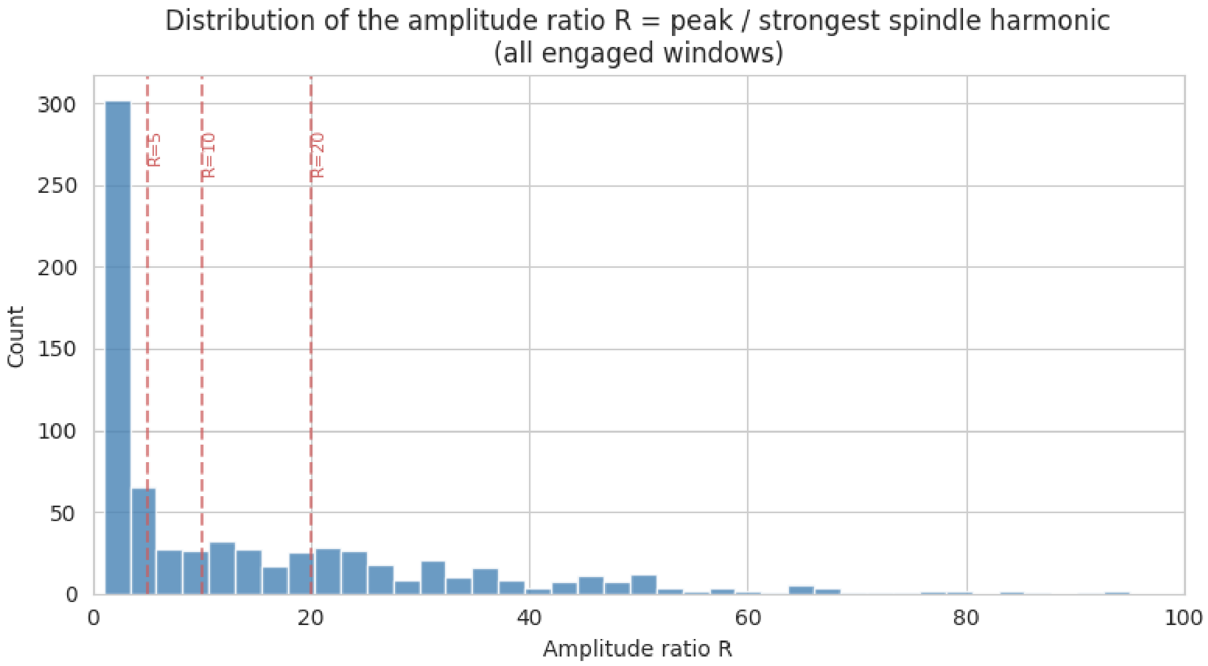
A formal patent criterion: chatter frequency is not an integer multiple of (cutting edges) x (spindle speed) -- for turning, not a spindle-rotation harmonic.

4 Persistence / monotonicity

Chatter grows and sustains once triggered, unlike a transient spike. MADSA (moving-average difference spectrum analysis) tracks this across successive windows.

A Physics-Informed Chatter Label

Amplitude ratio + harmonic exclusion + persistence, replacing an 85th-percentile cutoff



Label rule

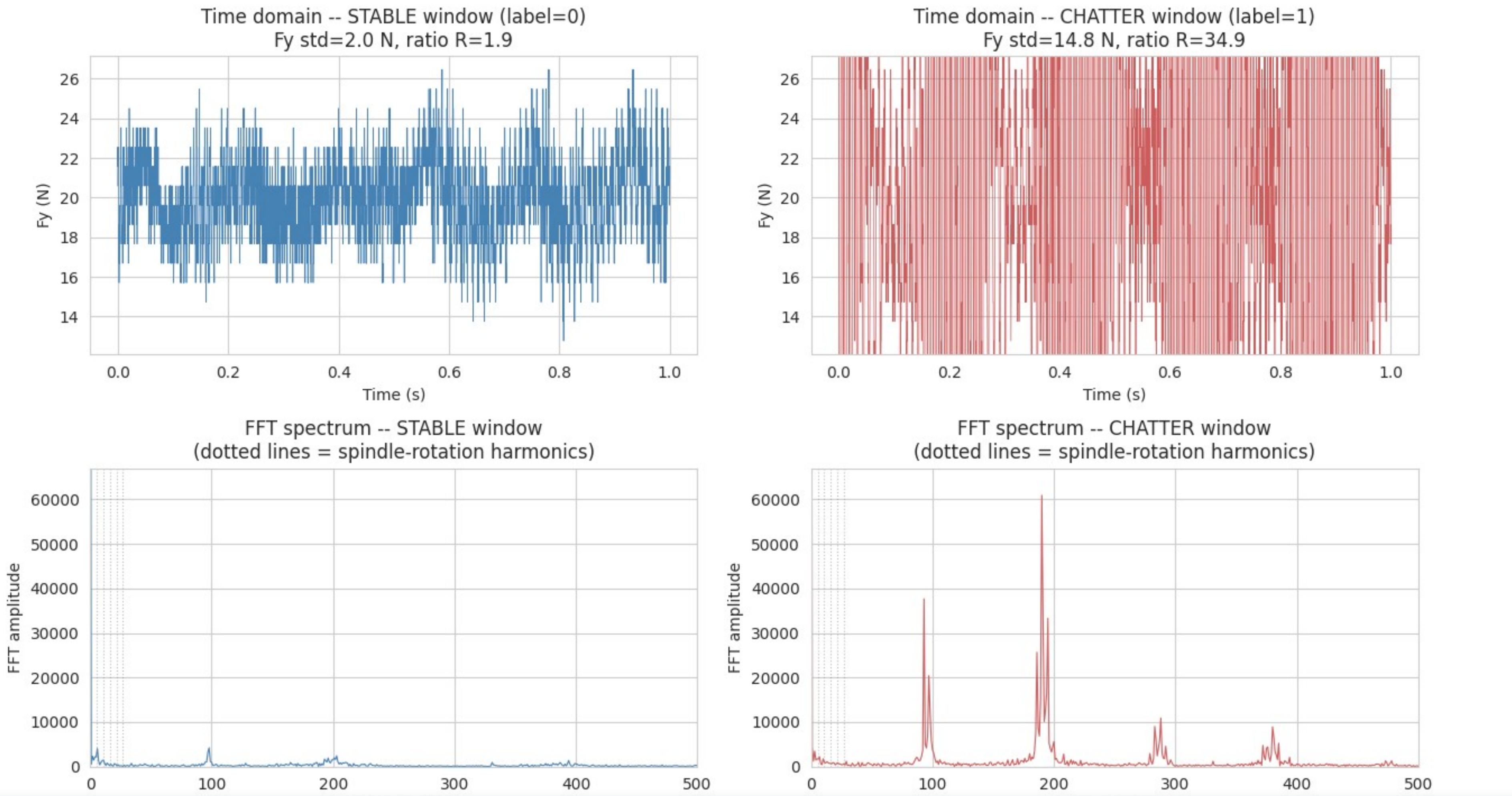
1. R = peak amplitude / strongest spindle-rotation harmonic in the same window
2. Chatter candidate if R > 10 AND peak is not itself a harmonic
3. Confirmed only if sustained across 2 consecutive overlapping windows

42.8% chatter fraction (vs. 15% fixed by the naive rule)

Agrees with the naive label on 71.7% of windows; where they disagree, the physics-informed label catches 207 real chatter windows the naive rule missed (vs. 2 in the opposite direction).

What Does This Actually Look Like on the Signal?

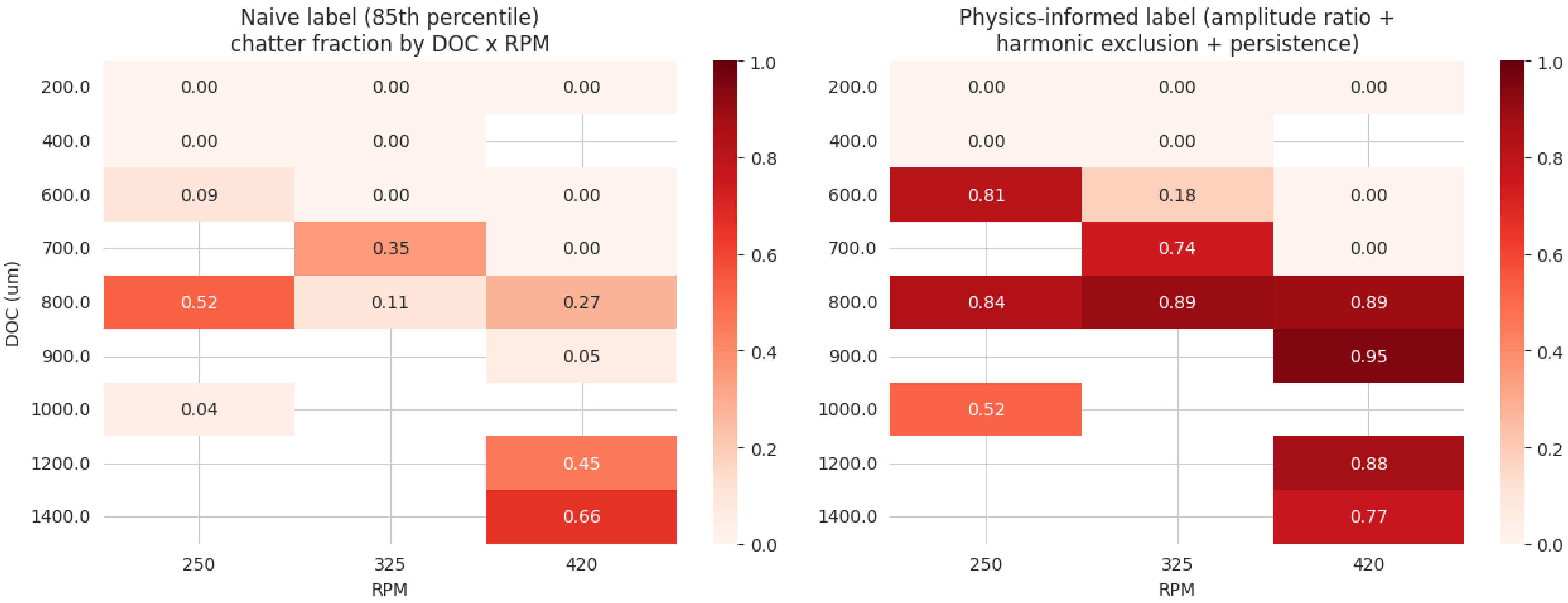
A real stable window vs. a real chatter-labeled window, same file -- so DOC and RPM are held constant



The stable window's spectrum (left) is broad and low. The chatter window's spectrum (right) has tall, distinct peaks (~190 Hz + harmonics) that dwarf the spindle-rotation harmonics (dotted grey lines) -- the "sudden extreme peak" signature directly visible in real data.

Validating the New Label Against Stability-Lobe Theory

The physics-informed label was never told DOC or RPM -- only each window's own force spectrum



The physics-informed label (right) shows a consistently high (75-95%) chatter fraction once DOC exceeds ~700 um, matching the sharp-threshold behavior expected from stability-lobe theory. The naive label (left) is patchier -- e.g. an unexplained dip to 5% at 900 um/420 RPM despite much higher-chatter neighboring conditions.

A Deliberate Check for Label Leakage

Before trusting the modeling results, the label's relationship to engineered features was checked directly

The issue found

The dimensionless "peakedness" features (crest, impulse, shape, clearance factors) correlate with the physics-informed label at 0.83-0.86 -- too high to be independent evidence. Both the label and these features are, structurally, different arithmetic operations on the same underlying signal property (peak concentration), just computed in different domains (time vs. frequency).

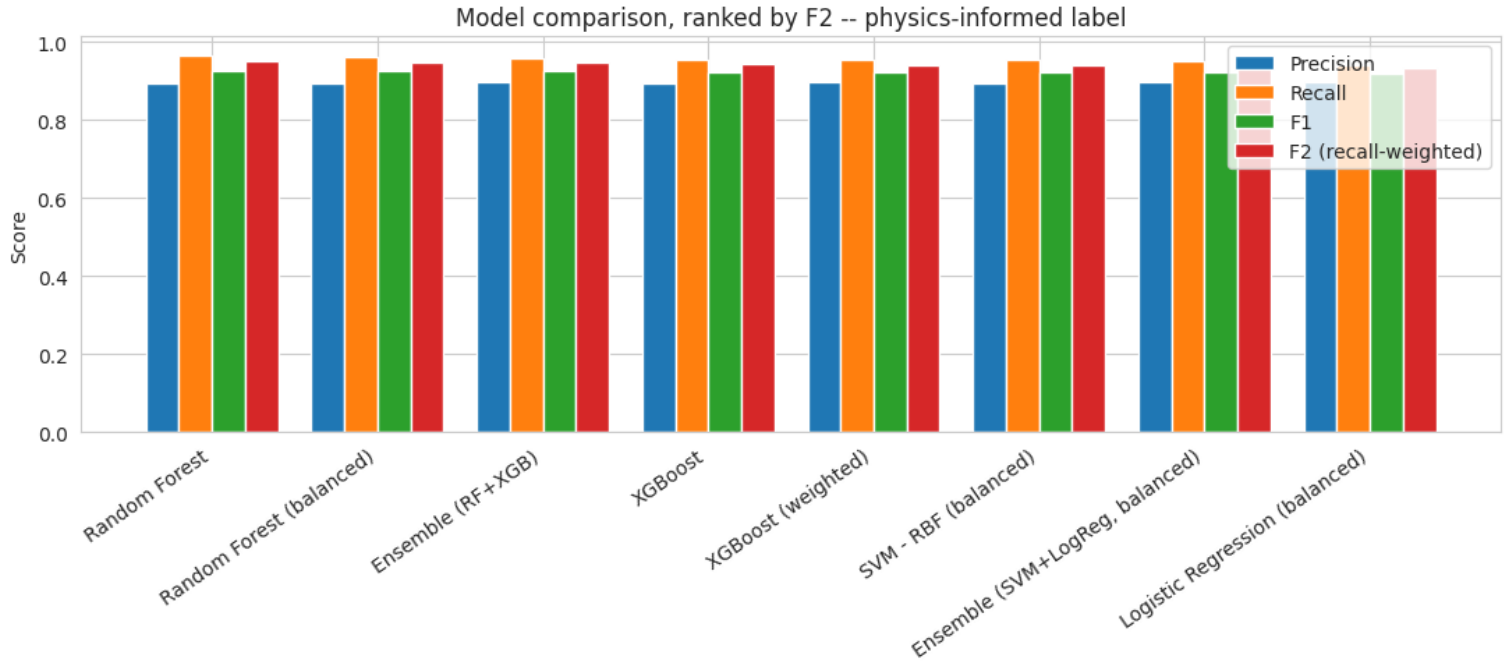
The fix

All four peakedness features (all three axes) were excluded from the candidate set, leaving standard deviation, range, RMS, energy, min/max, skewness, kurtosis, slope, and dominant frequency -- statistics that are conceptually distinct from a peak-concentration ratio. Final top-5 features: Fy_std, Fy_max, Fy_range, Fz_min, Fz_fft_peak.

This kind of construct-validity check matters more than the headline accuracy number -- it is what makes the reported results defensible.

Results

Ranked by F2 (recall weighted 2x precision) -- missing a chatter event costs more than a false alarm



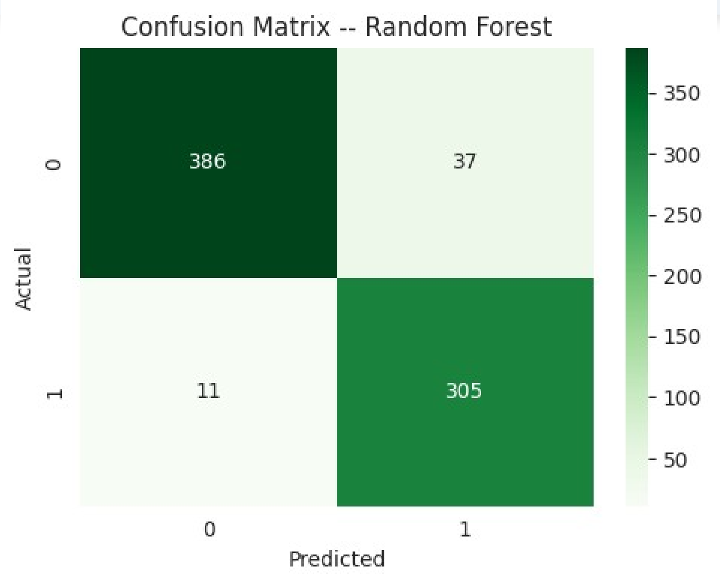
All 8 models cluster near 0.9+ F1 -- consistent with a genuinely cleaner label being the driver of higher scores, not one special model. All metrics are pooled out-of-fold, matching the confusion matrix exactly.

Final model

Random Forest

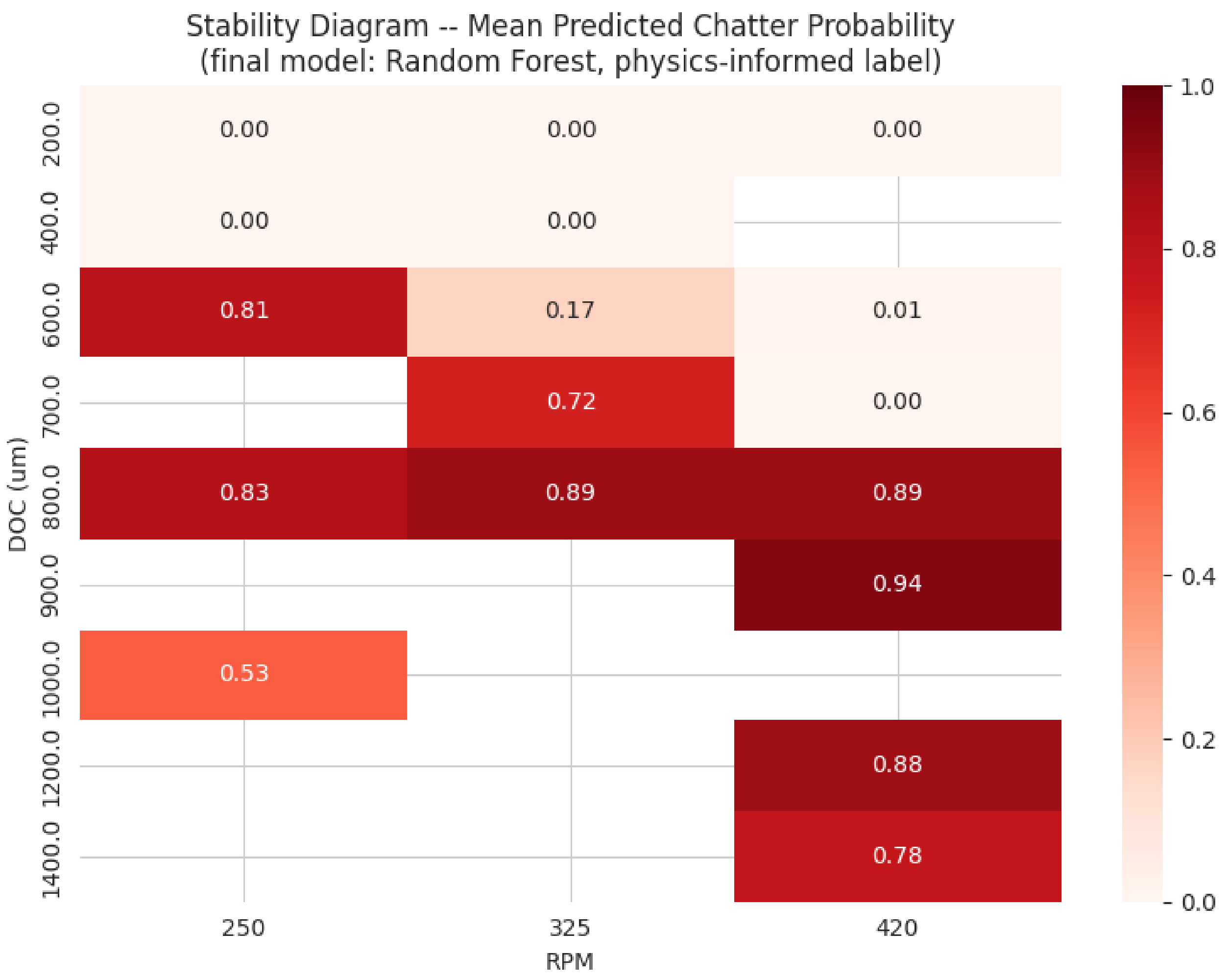
(top-10 features, physics-informed label)

Precision	Recall	F2
0.89	**0.97**	**0.95**



Stability Diagram — Chatter Probability by DOC x RPM

Final model refit on all data, aggregated by cutting condition, for process planning



Reading the diagram

- Chatter probability is 0% at DOC <= 400 um across all spindle speeds.
- Rises sharply to 72-94% at DOC 700-900 um.
- Stays elevated (77-88%) at the highest depths tested (1200-1400 um).
- This diagram reflects what the model learned from the label -- it summarizes the label's own pattern, not an independently re-derived stability boundary (see Limitations).

Conclusion & Limitations

Key contributions

- Automatic engagement detection (+ charge-drift guard), validated by ablation study.
- A physics-informed chatter label built from 3 literature-grounded criteria, replacing a naive percentile cutoff.
- Label validated against stability-lobe theory: clean, monotonic DOC relationship.
- A deliberate leakage check caught and corrected a construct-validity issue before trusting results.
- Final model (Random Forest): precision 0.89, recall 0.97, F2 0.95, selected to reflect the real cost of missed chatter.

Limitations & next steps

- The label, however improved, is still derived from the same signal used to build the features -- not an independently measured ground truth (accelerometer, microphone, surface roughness).
- Threshold values (ratio>10, persistence=2 windows) were tuned on this dataset's own distribution.
- 17 independent cuts -- GroupKFold metrics carry real variance.
- Next: independent validation signal, threshold sensitivity analysis, leave-one-DOC-out evaluation.